A Case Study Report
on

"Case Study 3: Predicting Titanic Survival with Logistic Regression"

and

"Case Study 6: Comparative Analysis of Classification Algorithms for Bank Marketing Prediction"

for the course

Fundamentals of Data Science

Submitted in partial fulfillment of the requirements for the award of degree

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Abstract

This report presents two case studies from the Fundamentals of Data Science course that apply supervised machine learning to real datasets. Case Study 3 predicts whether a Titanic passenger survived using logistic regression. After exploring the data, four new features were created (Title, FamilySize, IsAlone and HasCabin) and the model was tuned with GridSearchCV, which raised test accuracy from 0.8045 to 0.8324 and ROC-AUC from 0.8435 to 0.8688. The coefficients and statsmodels p-values show that being a man, travelling in a lower class, a larger family and older age lowered the chance of survival, and the final model was deployed as a Streamlit web app.

Case Study 6 predicts whether a bank customer will subscribe to a term deposit after a telephone marketing campaign. Four classifiers (Logistic Regression, Decision Tree, K-Nearest Neighbours and Random Forest) were tuned and compared on the same stratified 80:20 split, with the call duration removed to avoid data leakage. Because only 11.27% of customers subscribed, the models were compared on precision, recall, F1-score and ROC-AUC rather than on accuracy alone. Random Forest gave the best balance (F1-score 0.5065, ROC-AUC 0.8134), and calling the top 20% of customers ranked by its score would reach 66.38% of all subscribers.

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1. INTRODUCTION

This report covers two case studies from the Fundamentals of Data Science course. Both are binary classification problems on real data, and both follow the same data science process: understanding and cleaning the data, exploring it, building models, evaluating them on unseen test data and interpreting the results.

1.1 Problem Definition including the significance and objective

Problem Definition

In **Case Study 3**, the task is to predict whether a passenger on the Titanic survived from basic details such as class, sex, age, family and fare, and to find out which of these factors mattered most. In **Case Study 6**, a Portuguese bank sells term deposits through telephone campaigns in which most calls fail. The task is to predict, before a call is made, which customers are likely to subscribe, compare four classification algorithms and recommend the most useful one for the bank. In both datasets one class is smaller than the other, so the choice of evaluation metrics matters as much as the choice of model.

Significance

- Shows how a simple, interpretable model such as logistic regression can explain which factors drive an outcome, not only predict it.
- Shows how preprocessing choices (missing values, encoding, scaling and removing leaked features) decide whether the results can be trusted.
- For the bank, a good model lets the marketing team call the most promising customers first, saving calls and finding more subscribers.
- Shows why accuracy can be misleading on imbalanced data and how precision, recall, F1-score and ROC-AUC give a fairer comparison.

Objectives

1. To explore both datasets with summary statistics and visualisations and describe the patterns found.
2. To handle missing and 'unknown' values, encode categorical variables and scale features without leaking test data.
3. To build, tune and evaluate a logistic regression model for Titanic survival, interpret its coefficients and deploy it with Streamlit.

4. To build, tune and compare Logistic Regression, Decision Tree, KNN and Random Forest classifiers on the bank marketing data.
5. To choose the best model using confusion matrices, precision, recall, F1-score and ROC-AUC, and give business recommendations.

1.2 Concepts used in Methodology

Data Preparation Concepts

- **Exploratory data analysis (EDA):** summary statistics, histograms, box plots, bar charts and correlation heatmaps to understand distributions and relationships.
- **Preprocessing pipeline:** missing values are imputed (median for numbers, most frequent value for categories), categories are one-hot encoded and numbers are scaled with StandardScaler. All steps sit inside a scikit-learn Pipeline fitted on the training data only, so nothing from the test set leaks into training [3].
- **Stratified 80:20 split:** the test set keeps the same class proportions as the full data (random_state = 42).
- **Feature engineering:** new variables are built from existing ones, for example Title and FamilySize in the Titanic data.

Machine Learning Concepts

- **Logistic Regression:** models the log-odds of the positive class as a linear combination of the features; exp(coefficient) gives the odds ratio.
- **Decision Tree:** splits the data with if-then rules on the features; depth and leaf size control overfitting.
- **K-Nearest Neighbours (KNN):** classifies a record by the majority class of the k most similar records in the training data.
- **Random Forest:** an ensemble of many decision trees trained on random samples of rows and features, whose votes are combined.
- **GridSearchCV:** tries every combination of hyperparameters in a grid using k-fold cross-validation on the training set and keeps the best one.

Evaluation Concepts

- **Confusion matrix** (TN, FP, FN, TP), **accuracy**, **precision** (share of predicted positives that are correct), **recall** (share of actual positives that are found), **F1-score** (harmonic mean of precision and recall) and **ROC-AUC** (how well the model ranks positives above negatives).

- **statsmodels Logit p-values** to test whether each coefficient differs significantly from zero at the 0.05 level.
- **Cumulative gains:** the share of all subscribers reached when only the top-scored customers are called.

1.3 Outline the results

Chapter 2 presents Case Study 3 and Chapter 3 presents Case Study 6. The main results are:

- **Case Study 3:** new features and tuning raised test accuracy from 0.8045 to 0.8324, recall from 0.6667 to 0.7681 and ROC-AUC from 0.8435 to 0.8688.
- Title (Mr), passenger class, family size, age and having a cabin were statistically significant predictors of survival, and the tuned model was deployed as a Streamlit app.
- **Case Study 6:** Random Forest was the best model (F1-score 0.5065, ROC-AUC 0.8134, recall 0.6336), catching 588 of the 928 subscribers in the test set.
- KNN had the highest accuracy (0.8950) but found only 278 subscribers, which is why accuracy alone was not used to choose the model.
- Calling the top 20% of customers ranked by the Random Forest score reaches 66.38% of all subscribers.

2. CASE STUDY 3: TITANIC SURVIVAL PREDICTION

2.1 Problem Statement

The aim is to predict whether a Titanic passenger survived using details like class, sex, age, family and fare, and to find out which factors mattered most. I also tried to improve a basic logistic regression by adding new features and tuning it, and I deployed the final model as a small web app.

2.2 Dataset Description

The dataset (titanic.csv) was provided by the faculty [2]. It has 891 passengers and 12 columns. The target is Survived (1 = yes), and 38% of passengers survived. Missing values: Cabin 687 (77.10%), Age 177 (19.87%) and Embarked 2 (0.22%). Table 2.1 summarises the numeric features.

Table 2.1: Summary statistics of the Titanic numeric features

Feature	Count	Mean	Std	Min	Median	Max
Age	714	29.70	14.53	0.42	28.00	80.00
Fare	891	32.20	49.69	0.00	14.45	512.33
SibSp	891	0.52	1.10	0.00	0.00	8.00
Parch	891	0.38	0.81	0.00	0.00	6.00
Pclass	891	2.31	0.84	1.00	3.00	3.00

2.3 Methodology

Figure 2.1 shows the steps followed, from loading the data to deploying the model.

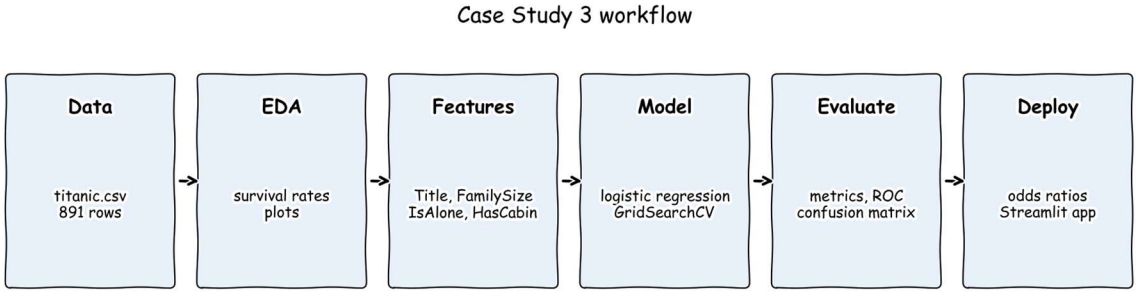


Fig 2.1: Workflow followed for Case Study 3

- **New features:** Title from the name (Mr, Mrs, Miss, Master, Rare), FamilySize = SibSp + Parch + 1, IsAlone and HasCabin.

- **Baseline model:** the 7 original features. **New model:** Pclass, Age, Fare, Embarked, Title, FamilySize, IsAlone and HasCabin (Sex is not repeated because Title already includes it).
- **Preprocessing inside a Pipeline:** median imputation and scaling for numbers, most-frequent imputation and one-hot encoding for categories, fitted on training data only. Stratified 80:20 split with random_state = 42.
- **Tuning:** GridSearchCV over C (0.001 to 100) and penalty L1/L2, 5-fold cross-validation, scored by ROC-AUC.
- **Metrics:** accuracy, precision, recall, F1-score and ROC-AUC, plus statsmodels p-values for the coefficients.

2.4 Statistical Findings

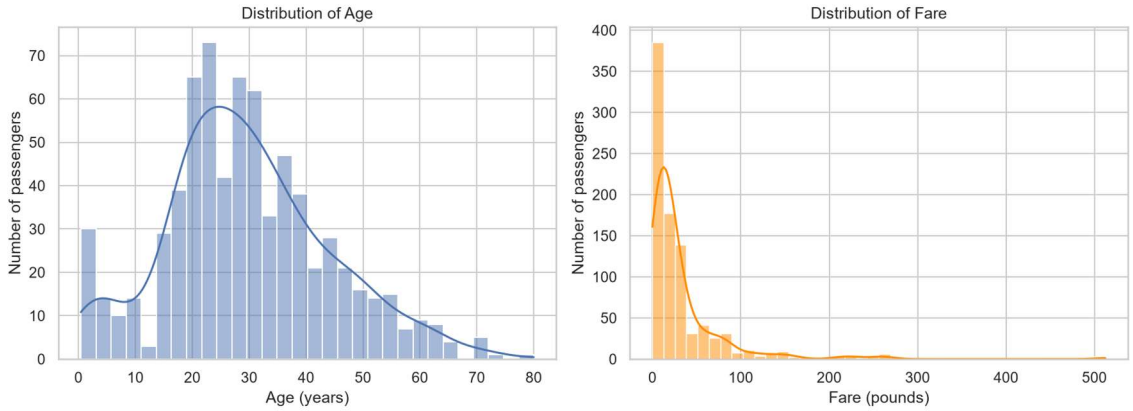


Fig 2.2: Distributions of passenger age and fare

Most passengers were between 20 and 40 years old (median 28, mean 29.70), with a small group of young children under 5. Fare is strongly right-skewed: the median fare was 14.45 but the mean was 32.20, because a few passengers paid very high fares of up to 512.33.

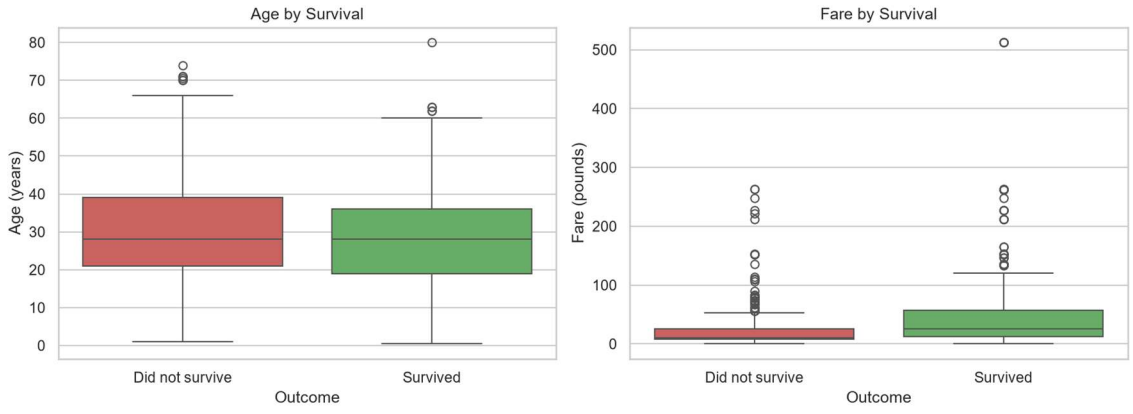


Fig 2.3: Age and fare by survival outcome

The median age is almost the same for survivors and non-survivors, so age on its own does not separate the two groups well. Survivors tended to pay higher fares, and both groups have many high-fare outliers, which come mostly from first-class tickets.

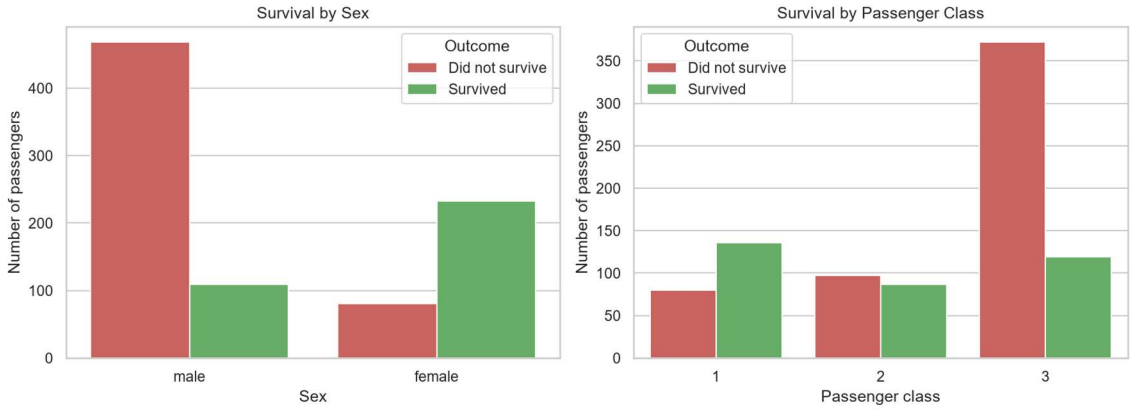


Fig 2.4: Survival rate by sex and by passenger class

Women survived much more than men (74.20% against 18.89%). Survival also falls with class, from 62.96% in 1st class to 24.24% in 3rd class.

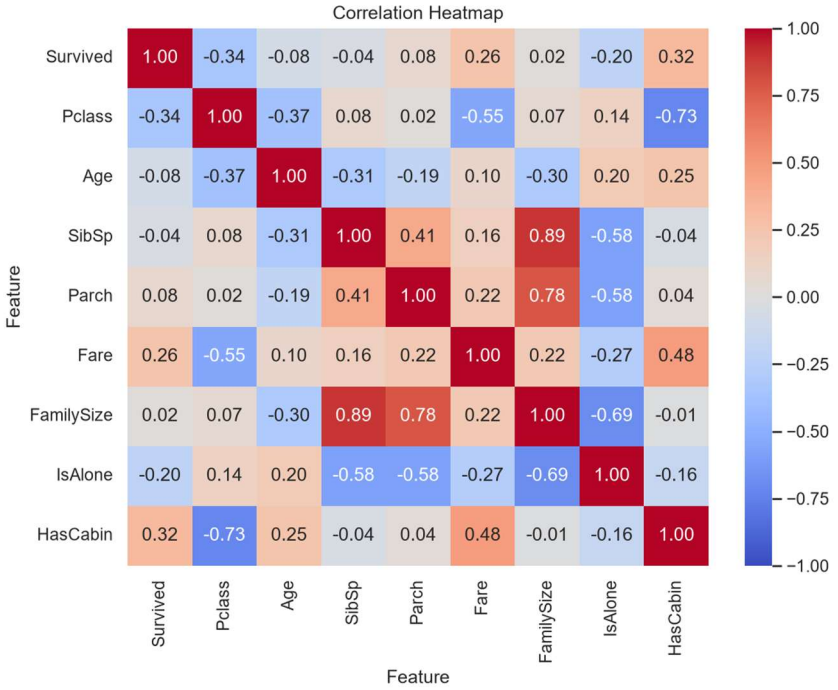


Fig 2.5: Correlation heatmap of the numeric and engineered features

Survival is most strongly correlated with Pclass (-0.338), HasCabin (0.317) and Fare (0.257), and negatively with IsAlone (-0.203). Age has almost no linear correlation with survival (-0.077). Pclass is strongly related to HasCabin (-0.726) and Fare (-0.549), so these features partly carry the same information. FamilySize is built from SibSp and Parch, which is why it

is highly correlated with them (0.891 and 0.783) and why SibSp and Parch were not used in the new model.

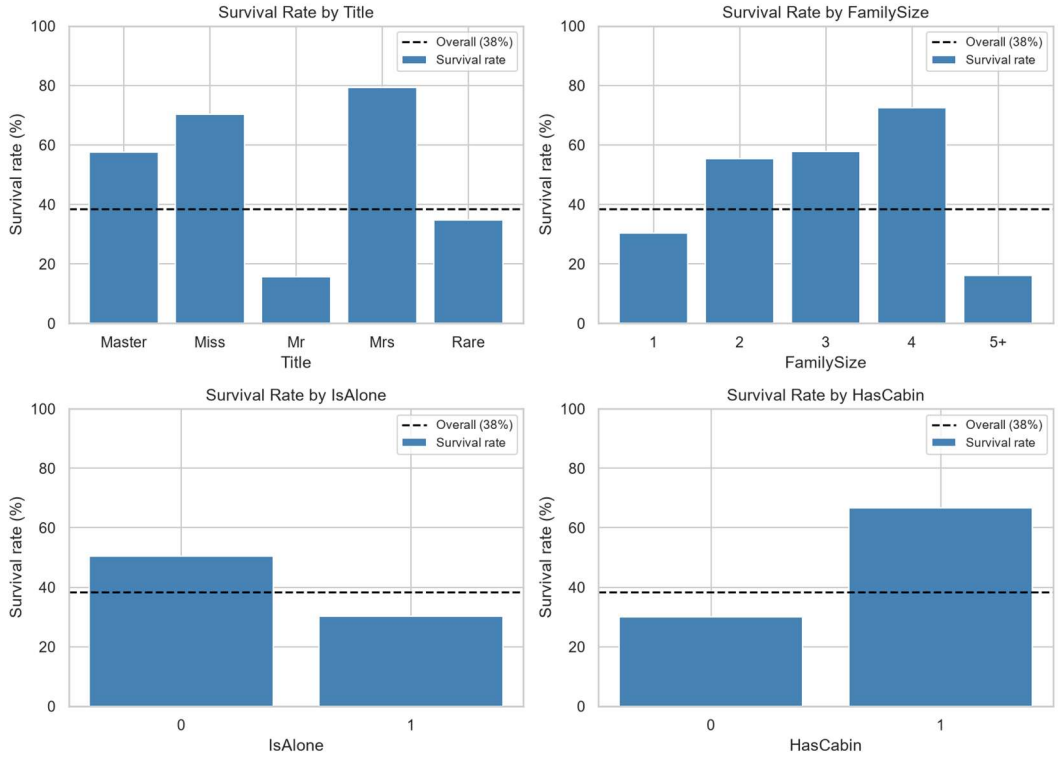


Fig 2.6: Survival rate for the engineered features

The new features show clear patterns. Men (Mr) survived at only 15.67%, but boys (Master) at 57.50% and married women (Mrs) at 79.37%. People travelling alone (30.35%) and large families of 5 or more (16.13%) did worse than families of four (72.41%). Passengers with a known cabin survived at 66.67% against 29.99%.

2.5 Model Evaluation

Table 2.2: Test set results (179 passengers)

Model	Accuracy	Precision	Recall	F1	ROC-AUC	TN/FP/FN/TP
Baseline (7 features)	0.8045	0.7931	0.6667	0.7244	0.8435	98/12/23/46
Tuned new model (C = 3, L2)	0.8324	0.7910	0.7681	0.7794	0.8688	96/14/16/53

Grid search picked C = 3 with L2 (cross-validated ROC-AUC 0.8721). 5-fold cross-validated accuracy went from 0.7949 (baseline) to 0.8259 (tuned). On the test set the tuned model is better on accuracy (0.8045 to 0.8324), recall (0.6667 to 0.7681), F1-score and ROC-AUC, and it finds 53 of the 69 survivors instead of 46.

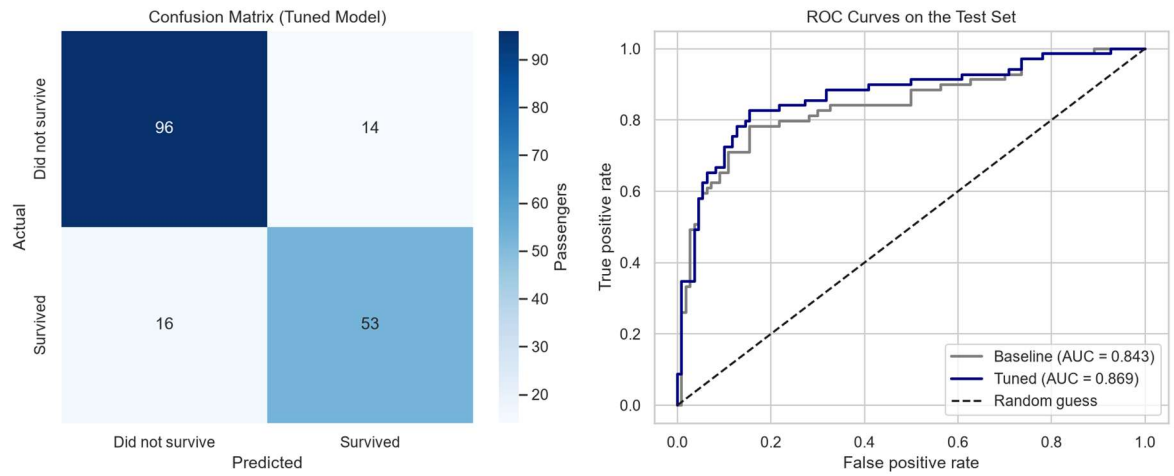


Fig 2.7: Confusion matrix of the tuned model and ROC curves of both models

A false positive (14) is a passenger predicted to survive who actually died, and a false negative (16) is a survivor the model missed. Precision 0.7910 means the model is right about 79% of the time when it says 'survived', and recall 0.7681 means it finds about 77% of real survivors. ROC-AUC 0.8688 shows that it ranks survivors above non-survivors well.

2.6 Interpretation and Deployment

Table 2.3: Coefficients of the tuned model with statsmodels p-values

Feature	Coefficient	Odds ratio	p-value	Significant
Title_Mr	-2.5167	0.0807	< 0.001	Yes
Title_Rare	-1.5152	0.2198	0.0112	Yes
Title_Mrs	1.0018	2.7232	0.2919	No
HasCabin	0.9639	2.6219	0.0056	Yes
FamilySize	-0.6704	0.5115	< 0.001	Yes
Pclass	-0.6028	0.5473	< 0.001	Yes
Age	-0.4112	0.6629	0.0052	Yes
Embarked_Q	0.3278	1.3879	0.5002	No
Embarked_S	-0.3120	0.7320	0.2356	No
IsAlone	-0.2963	0.7435	0.3483	No
Title_Miss	0.2185	1.2442	0.8126	No
Fare	0.1096	1.1158	0.4496	No

Title_Mr has the biggest effect: a man's odds of survival were about 8% of a boy's (Master). A known cabin multiplies the odds by 2.62. Larger families, lower class and older age lower the odds. The statsmodels fit is significant overall (pseudo R-squared 0.3892). Significant at 0.05: Pclass, Age, FamilySize, HasCabin, Title_Mr and Title_Rare. Not significant: Fare, IsAlone, Embarked_Q, Embarked_S, Title_Miss and Title_Mrs.

The tuned pipeline is saved with joblib (3/model/titanic_logreg.joblib). The Streamlit app (3/app.py) takes the passenger details, works out FamilySize and IsAlone, and shows the survival probability and the predicted class. It runs with streamlit run 3/app.py.

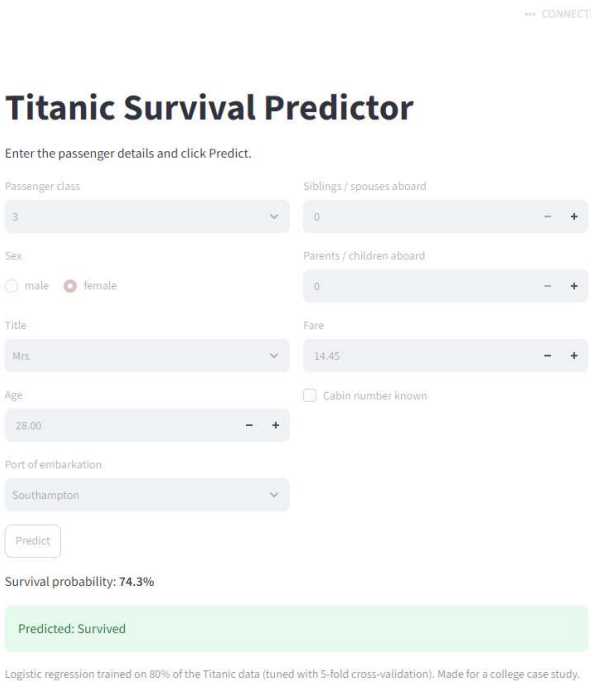


Fig 2.8: Streamlit app for survival prediction

2.7 Conclusion

Being a man, being in a lower class and family size were the main factors in survival. The new features and tuning raised test accuracy from 0.8045 to 0.8324 and ROC-AUC from 0.8435 to 0.8688. Limitations: the test set is small (179 passengers), about 20% of ages had to be filled in, and the data comes from one historical event.

3. CASE STUDY 6: BANK MARKETING CLASSIFICATION

3.1 Problem Statement

A Portuguese bank calls customers to sell term deposits, and most calls fail. The aim is to predict before calling which customers will subscribe, compare four classifiers (Logistic Regression, Decision Tree, KNN and Random Forest) and recommend the best one for the bank.

3.2 Dataset Description

I used bank-additional-full.csv from the UCI repository [1] (41,188 rows, 21 columns, May 2008 to November 2010). I chose it over bank-full.csv because it is newer and has five economic indicators. After removing 12 duplicates, 4,639 customers (11.27%) subscribed and 36,537 (88.73%) did not, so the data is very imbalanced. There are no NaN values, but 'unknown' appears in several columns, most often in default (8,597, 20.87%). Table 3.1 summarises the main numeric features.

Table 3.1: Summary statistics of the bank marketing numeric features

Feature	Mean	Std	Min	Median	Max
age	40.024	10.421	17.000	38.000	98.000
campaign	2.568	2.770	1.000	2.000	56.000
pdays	962.475	186.911	0.000	999.000	999.000
previous	0.173	0.495	0.000	0.000	7.000
euribor3m	3.621	1.734	0.634	4.857	5.045
nr.employed	5167.036	72.252	4963.600	5191.000	5228.100

3.3 Methodology

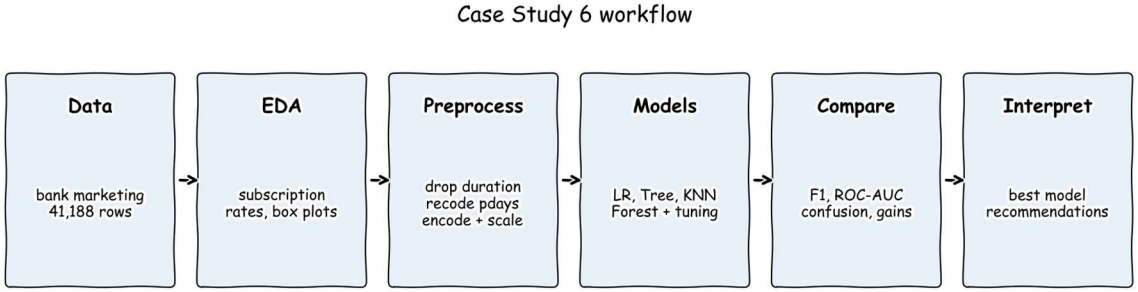


Fig 3.1: Workflow followed for Case Study 6

- **duration is removed** from the models because it is only known after the call (data leakage).

- **pdays = 999** means no previous contact, so it is replaced by a flag (previously_contacted) and the real number of days (pdays_contacted).
- **'unknown' is kept as its own category.** Numbers are scaled and categories one-hot encoded inside a Pipeline, with one stratified 80:20 split for all models.
- **Tuning:** each model is tuned with GridSearchCV (3-fold cross-validation, scored by F1). Logistic Regression, Decision Tree and Random Forest use class_weight = 'balanced'; KNN has no such option.

Table 3.2: Best settings from the grid search

Model	Best parameters	CV F1
Logistic Regression	C = 10	0.4498
Decision Tree	max_depth = 4, min_samples_leaf = 20	0.4581
KNN	n_neighbors = 5, weights = uniform	0.3730
Random Forest	max_depth = 12, min_samples_leaf = 1, n_estimators = 200	0.4835

3.4 Statistical Findings



Fig 3.2: Distribution of the target variable (term deposit subscription)

Only 4,639 of 41,176 customers (11.27%) subscribed. A model that always predicts 'no' would already be 88.73% accurate, so accuracy is not a useful way to compare the models on this data.

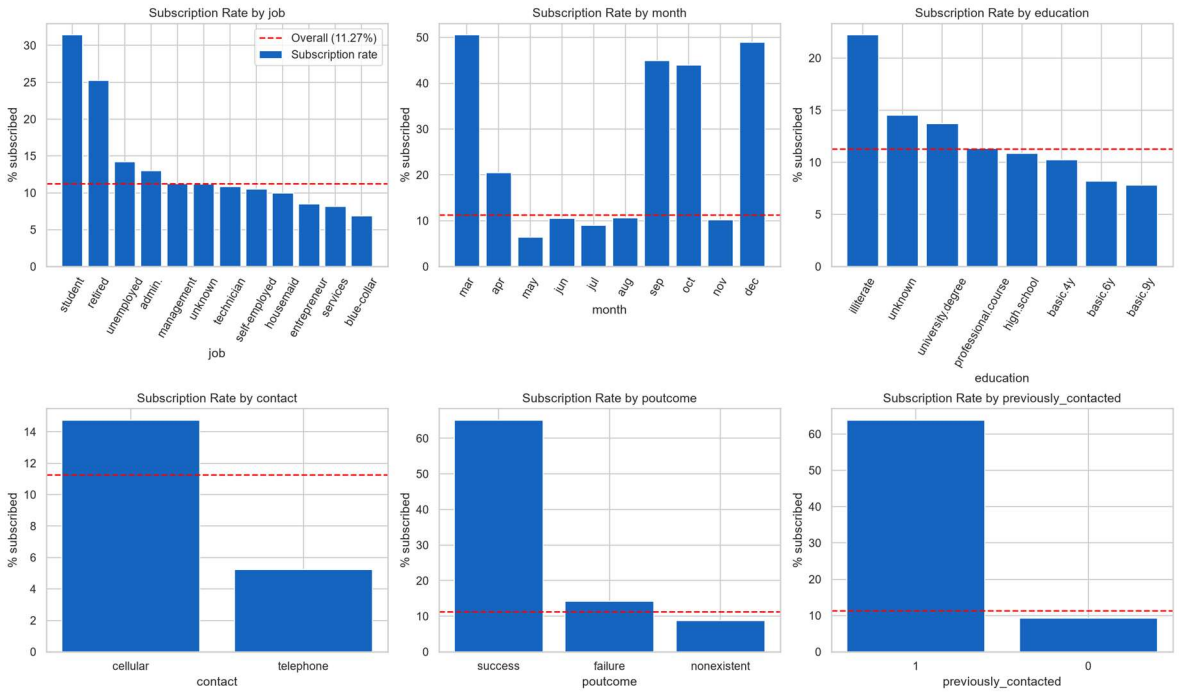


Fig 3.3: Subscription rate across customer and campaign attributes

- Customers contacted before subscribe at 63.83%, against 9.26% for the rest, and a previous campaign success gives the highest rate of all (65.11%).
- May had the most calls (13,767) but the lowest rate (6.44%), while March, September, October and December had rates above 40% from far fewer calls.
- Cellular contact (14.74%) beats telephone (5.23%).
- Students (31.43%) and retired customers (25.26%) subscribe the most, and blue-collar workers the least (6.90%).

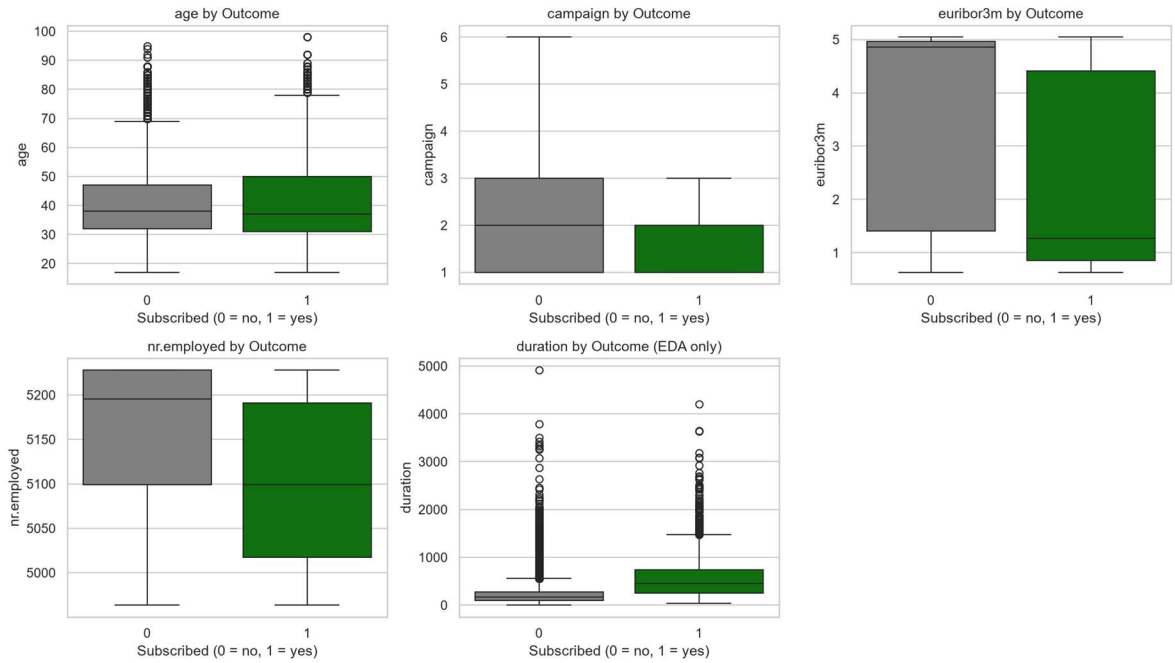


Fig 3.4: Numeric variables against subscription outcome

Interest rates matter: the median Euribor rate was 1.266 for subscribers and 4.857 for non-subscribers, and the median number employed was also lower for subscribers (5099.1 against 5195.8). Subscribers received slightly fewer calls in the campaign (mean 2.052 against 2.633), while age is almost the same in both groups (median 37 against 38). Calls with subscribers lasted much longer (median duration 449 against 164 seconds), but duration is only known after the call, so it is shown for EDA only and is not used in the models.

3.5 Model Evaluation

Table 3.3: Test set results (8,236 customers)

Model	Accuracy	Precision	Recall	F1	ROC-AUC	Subscribers caught
Logistic Regression	0.8298	0.3583	0.6455	0.4608	0.8001	599 / 928
Decision Tree	0.8462	0.3862	0.6196	0.4758	0.7852	575 / 928
KNN	0.8950	0.5639	0.2996	0.3913	0.7407	278 / 928
Random Forest	0.8609	0.4218	0.6336	0.5065	0.8134	588 / 928

KNN has the highest accuracy (0.8950) but finds only 278 of 928 subscribers, so accuracy is not a good measure here. Logistic Regression has the highest recall (0.6455) but the lowest precision (0.3583). Random Forest has the best F1-score (0.5065) and ROC-AUC (0.8134).

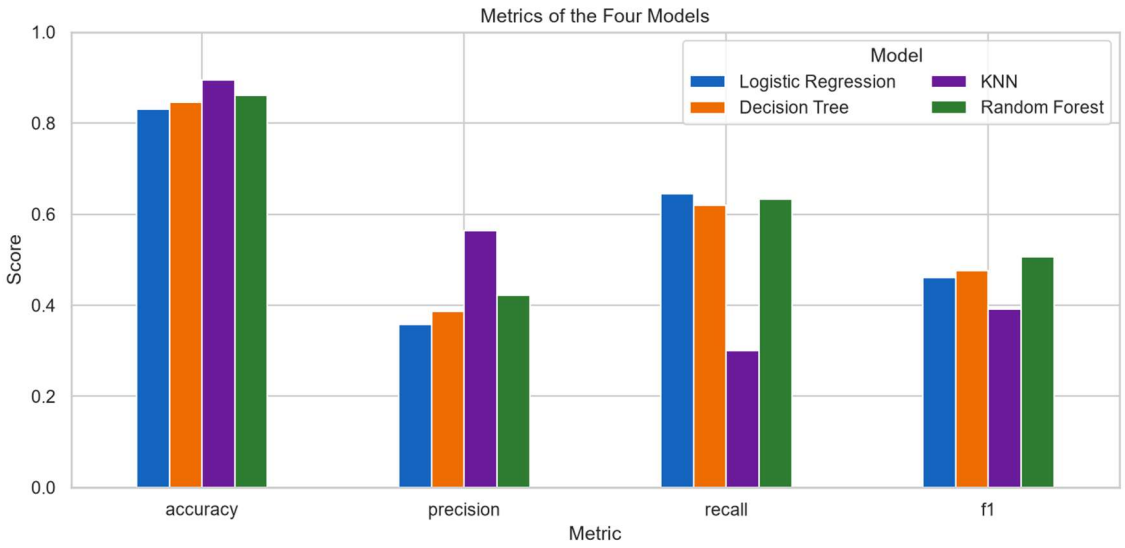


Fig 3.5: Accuracy, precision, recall and F1-score of the four models

The chart shows the trade-off between the models. KNN is highest on accuracy and precision but lowest on recall, while Logistic Regression, Decision Tree and Random Forest each find about 62% to 65% of subscribers. Random Forest has the highest F1-score, so it balances precision and recall best.

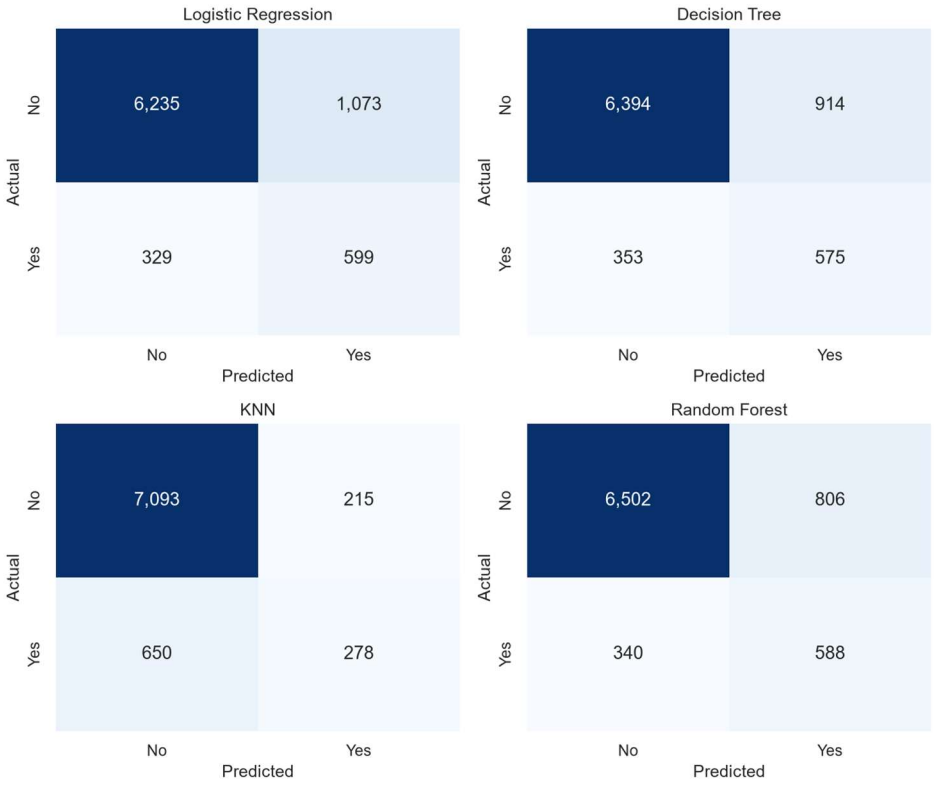


Fig 3.6: Confusion matrices of the four classifiers

A false negative is a customer who would have subscribed but is not called, so the bank loses a sale. A false positive is a wasted call. Random Forest has 340 false negatives and 806 false

positives, while Logistic Regression has 329 and 1,073. KNN makes the fewest false positives (215) but misses 650 subscribers.

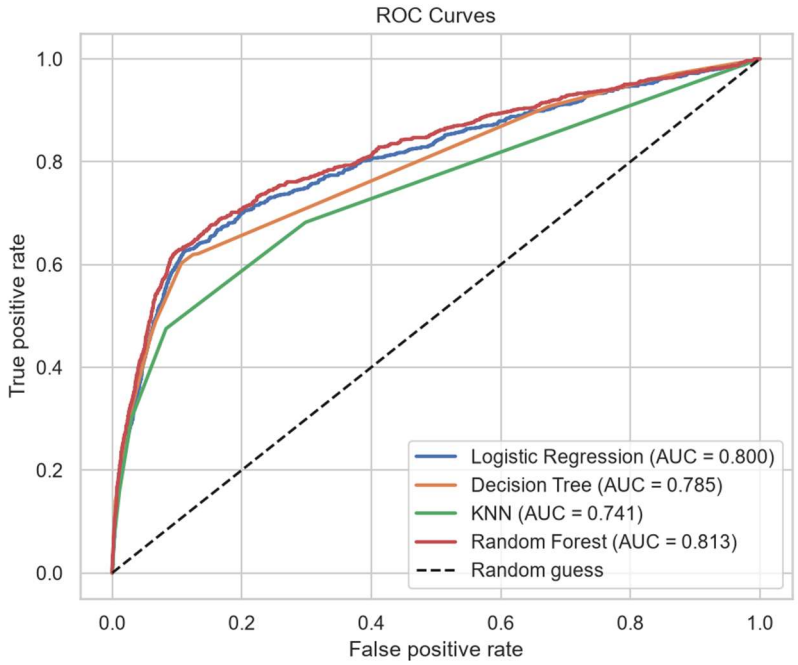


Fig 3.7: ROC curves of the four classifiers

All four curves lie well above the diagonal of random guessing. Random Forest has the largest area under the curve (0.8134), followed by Logistic Regression (0.8001), Decision Tree (0.7852) and KNN (0.7407).

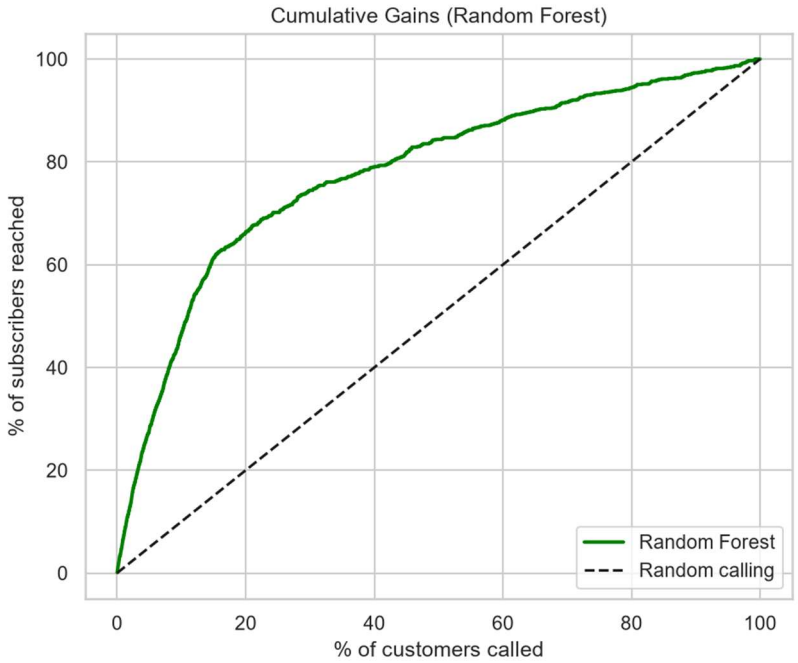


Fig 3.8: Cumulative gains of the Random Forest model

Ranking customers by the Random Forest score and calling the top 10% reaches 46.55% of subscribers, the top 20% reaches 66.38%, the top 30% reaches 74.46% and the top 50% reaches 84.38%. In the saved results for all four models, Random Forest is slightly ahead at every one of these levels.

3.6 Interpretation and Business Recommendations

Random Forest is the best model. It has the best F1-score and ROC-AUC, and compared with Logistic Regression it finds only 11 fewer subscribers but makes 267 fewer wasted calls.

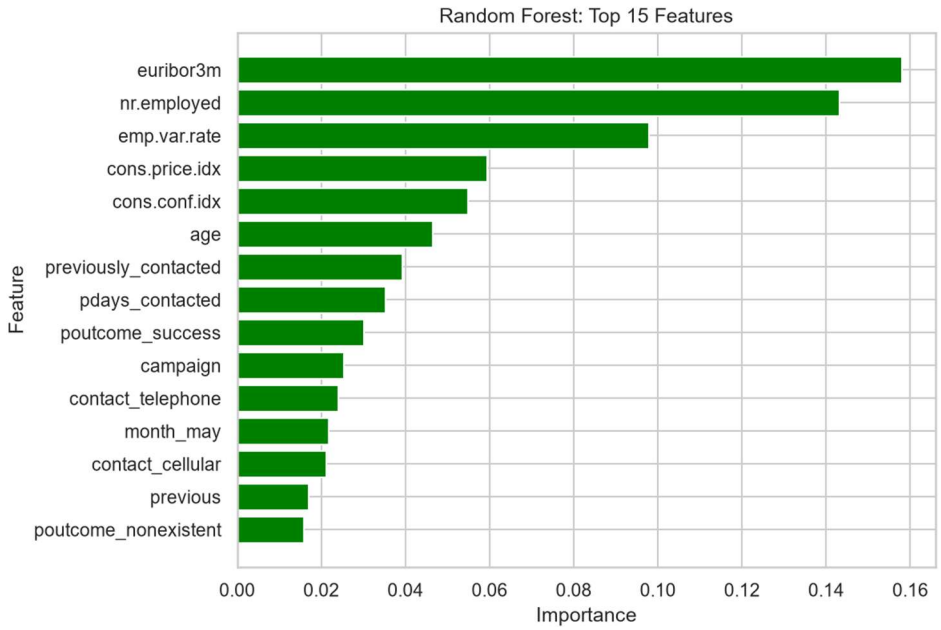


Fig 3.9: Top 15 Random Forest feature importances

The five economic features hold about 51% of the Random Forest's feature importance, led by euribor3m (0.1581) and nr.employed (0.1431). Among the customer and campaign features, age, previous contact and a successful previous outcome matter most. Based on these results, the bank should:

- Call customers in order of their Random Forest score instead of calling everyone.
- Focus on customers contacted before, students and retired customers.
- Use mobile phones, and run larger campaigns when interest rates are low.
- Not call the same customer too many times.

3.7 Conclusion and Limitations

After tuning, Random Forest gave the best balance (F1-score 0.5065, ROC-AUC 0.8134, recall 0.6336). Limitations: the data is from 2008 to 2010, so the economic effects may be different

today; removing duration keeps the model realistic but limits precision; all results use the 0.5 threshold; and the grids were small with a single test split.

4. CONCLUSION and References

4.1 Conclusion

Both case studies show that careful preprocessing and the right evaluation metrics matter as much as the choice of algorithm. In Case Study 3, a logistic regression model with engineered features (Title, FamilySize, IsAlone and HasCabin) and a tuned regularisation strength predicted Titanic survival with 0.8324 accuracy and 0.8688 ROC-AUC. Its coefficients gave a clear explanation: being a man, travelling in a lower class, a larger family and older age reduced the odds of survival. The model was deployed as a Streamlit app.

In Case Study 6, four tuned classifiers were compared on the same split with the leaked duration feature removed. Because only 11.27% of customers subscribed, accuracy was misleading: KNN had the highest accuracy but missed most subscribers. Random Forest gave the best balance of precision and recall (F1-score 0.5065, ROC-AUC 0.8134), and ranking customers by its score would let the bank reach about two thirds of subscribers by calling only the top 20% of customers.

4.2 References

[1] S. Moro, P. Cortez and P. Rita, “A Data-Driven Approach to Predict the Success of Bank Telemarketing,” Decision Support Systems, 2014, doi: 10.1016/j.dss.2014.03.001. Dataset: Bank Marketing, UCI Machine Learning Repository.

[2] Titanic passenger dataset (titanic.csv), provided by the course faculty.

[3] F. Pedregosa et al., “Scikit-learn: Machine Learning in Python,” Journal of Machine Learning Research, vol. 12, pp. 2825-2830, 2011.

[4] Python libraries: pandas, NumPy, Matplotlib, seaborn, statsmodels, joblib and Streamlit.

[5] GitHub repository: https://github.com/Nikhil-Kotte/FDS_CASE_STUDY.git